

GROUP10 | The Compounding Effect of Sadness & Alcohol on Youth Smoking

Research Question: Does psychological sadness have a compounding effect with substance dependency on adolescent smoking?

1 Study Framework

Group Definition

SadOrHopeless_Recoded (Binary)

Exposed Group (1): Felt sad/hopeless in past 12 months

Comparison Group (0): Did not feel sad

Response Variable:

CurrentCigaretteUse_Recoded (Binary)

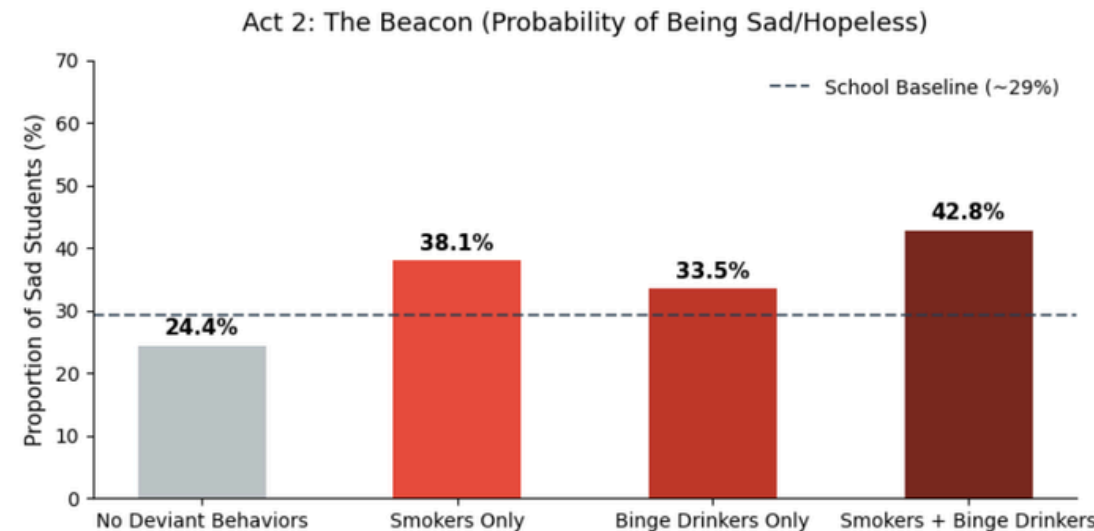
Success (1): Smoked ≥ 1 day in past 30 days

Failure (0): Smoked 0 days

2 Statistical Method

Method: Two-Proportion Z-Test

Assumption Check: Robustly passed the Success/Failure condition (All cell counts far exceed 10; Sample size $n = 13,174$).



3 Key Result

TOTAL SAMPLE (n)

13,174

Z-STATISTIC (p-value)

15.05 ($p < 0.001$)

COMPOUND RISK

7.8 x Higher

4 Conclusion

1. Shattering Rebellion Stereotypes

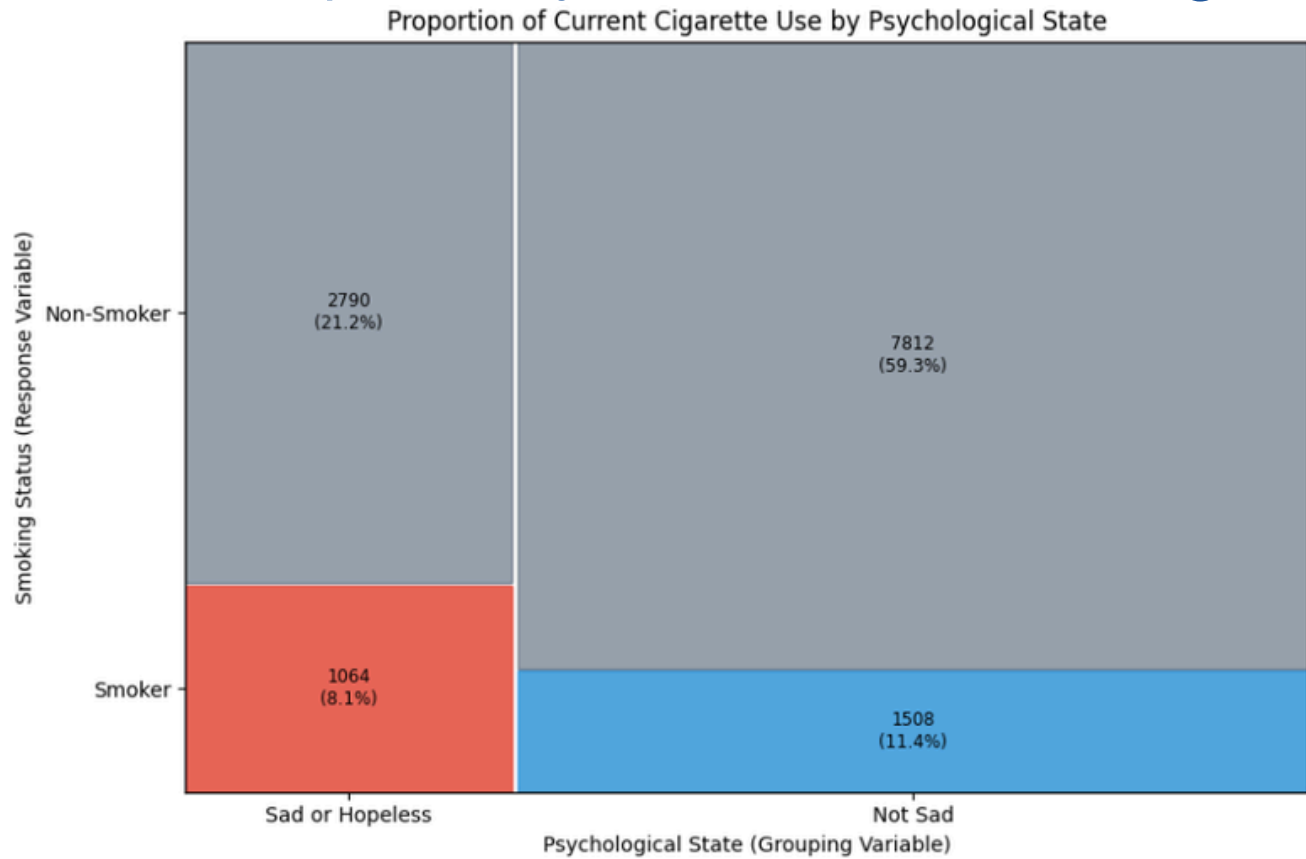
Pure rebellion accounts for only 20.1% of student smokers. The vast majority carry hidden, profound psychological or substance dependency burdens.

2. The Compounding Trap

Psychological sadness alone increases smoking rates to 27.6%. When combined with binge drinking, the rate of severe depression among smokers peaks at 42.8% (See Act 2).

3. Synergistic Risk & Integrated Policy

The overlay of sadness and alcohol dependency rockets the tobacco risk to 7.8x higher than the normal baseline (7.0%). School campaigns must pivot from behavior isolation to integrated models combining Tobacco Cessation with Student Mental Health Support.



27.6% of students in the "Sad or Hopeless" group are smokers, compared to only 16.18% in the "Not Sad" group.